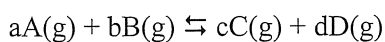


Set 3: Equilibrium constant expressions

Haber used equilibrium constant expressions to predict the concentration conditions that would give the greatest yield of ammonia when he investigated a method for producing it at a commercial level.

Equilibrium constant expressions are mathematic representations of the relationship between the concentrations of the products and reactants in a system at equilibrium. They are based on the equation for the reaction taking place. The value of the expression, the equilibrium constant, is constant for a given temperature.

Consider the following general equation for a system of gases in equilibrium:



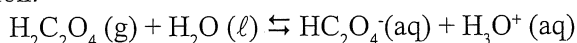
Where A, B, C and D are substances involved and a, b, c and d are the relative number of moles of each substance. The equilibrium constant expression for this reaction is written as:

$$K = \frac{[\text{Products}]}{[\text{Reactants}]} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

As the equilibrium constant expression uses concentrations, we do not include solids and liquids in the expression.

Example

Write the equilibrium constant expression for the ionisation of oxalic acid in water given the following equation:



As water is a liquid, it will not be included in the expression, so K will be:

$$K = \frac{[HC_2O_4^-][H_3O^+]}{[H_2C_2O_4]}$$

Set 3: Exercises

Write equilibrium constant expressions for the following equations.

1. $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$
2. $2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g)$
3. $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$
4. $Ag_2CrO_4(s) \rightleftharpoons 2Ag^+(aq) + CrO_4^{2-}(aq)$
5. $CO_2(aq) + H_2O(l) \rightleftharpoons HCO_3^-(aq) + H^+(aq)$

Notes

Set 3: Equilibrium constant expressions

Notes

6. $\text{HSO}_4^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{SO}_4^{2-}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
7. $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons 2\text{CrO}_4^{2-}(\text{aq}) + 2\text{H}^+(\text{aq})$
8. $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
9. $\text{HCO}_3^-(\text{aq}) + \text{NH}_3(\text{aq}) \rightleftharpoons \text{CO}_3^{2-}(\text{aq}) + \text{NH}_4^+(\text{aq})$
10. $\text{CaO}(\text{s}) + \text{SO}_3(\text{g}) \rightleftharpoons \text{CaSO}_4(\text{s})$
11. $2\text{NaHCO}_3(\text{s}) \rightleftharpoons \text{Na}_2\text{CO}_3(\text{s}) + \text{H}_2\text{O}(\text{g}) + \text{CO}_2(\text{g})$
12. $2\text{Hg}(\ell) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{Hg}_2\text{Cl}_2(\text{s})$
13. $\text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\ell) + \text{CO}_2(\text{aq}) \rightleftharpoons \text{Ca}^{2+}(\text{aq}) + 2\text{HCO}_3^-(\text{aq})$
14. $\text{CaCl}_2(\text{s}) + \text{H}_2\text{O}(\text{g}) + \text{CO}_2(\text{g}) \rightleftharpoons \text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{g})$

ANSWERS

Properties and Reactions

Set 1: Gases

- (i) 288 K (ii) 390 K (iii) 168 K
 - (i) 25 °C (ii) 200 °C (iii) -161 °C
- (a) 2.17×10^{-2} mol (b) 3.38×10^{-2} mol (c) 1.46 mol
- (a) 93.7 g mol^{-1} (b) 11.0 g mol^{-1} (c) 92.7 g mol^{-1}
- (a) 28.8 g mol^{-1} (b) C_2H_4 (ethene)
- 0.108 g
- (i) $\text{X} + 2\text{H}^+ \rightarrow \text{X}^{2+} + \text{H}_2$ (ii) $2\text{X} + 6\text{H}^+ \rightarrow 2\text{X}^{3+} + 3\text{H}_2$
 - (i) 22.7 g mol^{-1} (ii) 45.4 g mol^{-1} (iii) 68.1 g mol^{-1}
 - Sodium, as closest molar mass is Na.

Set 2: Limiting Reagents

- (a) NaHCO_3 (b) 8.51×10^{-2} L
- (a) CO (b) 1.00 L (c) 1.00 L CO_2 , 0.500 L O_2 , no CO
- (a) $\text{V}(\text{CO}_2) = 4.00 \times 10^{-2}$ L, $\text{V}(\text{H}_2\text{O}) = 4.00 \times 10^{-2}$ L, $\text{V}(\text{O}_2) = 2.00 \times 10^{-2}$ L, $\text{V}(\text{C}_2\text{H}_4) = 0$
 - (b) 0.100 L
- (a) Cu (b) 1.02 L
- 238 kPa
- 16.0 L
- 133 kPa
- 6.01×10^5 g or 0.601 tonne
- (a) KNO_3 (b) 15.0 g (c) 1.81×10^3 kPa

Set 3: Equilibrium constant expressions

- $$K = \frac{[\text{H}_2\text{O}]^2}{[\text{H}_2]^2[\text{O}_2]}$$
- $$K = \frac{[\text{N}_2]^2[\text{O}_2]}{[\text{N}_2\text{O}]^2}$$
- $$K = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2}$$
- $$K = [\text{Ag}^+]^2[\text{CrO}_4^{2-}]$$
- $$K = \frac{[\text{HCO}_3^-][\text{H}^+]}{[\text{CO}_2]}$$
- $$K = \frac{[\text{SO}_4^{2-}][\text{H}_3\text{O}^+]}{[\text{HSO}_4^-]}$$
- $$K = \frac{[\text{CrO}_4^{2-}]^2[\text{H}^+]^2}{[\text{Cr}_2\text{O}_7^{2-}]}$$
- $$K = [\text{CO}_2]$$
- $$K = \frac{[\text{CO}_3^{2-}][\text{NH}_4^+]}{[\text{HCO}_3^-][\text{NH}_3]}$$
- $$K = \frac{1}{[\text{SO}_3]}$$
- $$K = [\text{H}_2\text{O}][\text{CO}_2]$$
- $$K = \frac{1}{[\text{Cl}_2]}$$
- $$K = \frac{[\text{Ca}^{2+}][\text{HCO}_3^-]^2}{[\text{CO}_2]}$$
- $$K = \frac{[\text{CO}_3^{2-}][\text{NH}_4^+]}{[\text{HCO}_3^-][\text{NH}_3]}$$

Set 4: Application of Le Chatelier's Principle

- (a) $K = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$
 - (b) (i) decrease (ii) increase (iii) increase (iv) no change